Naledi Specialist Physicians (Rustenburg)

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1. Client Requirements Analysis

1.1 Client Business Understanding

Naledi Specialist Physicians is a healthcare organisation based in Rustenburg. The organisation requires a computer network that provides reliable connectivity between users, network devices and other required systems. Since the organisation operates in the healthcare industry, customer records are considered confidential and access to these records must be controlled.

The network must be designed and simulated using Cisco Packet Tracer. The design must use the assigned 10.12.0.0/16 addressing block and must demonstrate the assigned networking challenge, which is Spanning Tree Protocol (STP) for loop prevention and root design.

The network must also accommodate CR7, which requires four CCTV cameras to be added to the network and their traffic to be segmented.

1.2 Current Network Situation and Design Need

The client's current network requires improvement in terms of organisation, security, connectivity and reliability. The main issues and design needs identified are:

Current Network Situation / Problems:

- The network requires a more structured design to support the organisation's different users and devices.
- Confidential customer records require controlled access to prevent unauthorised users from accessing sensitive information.
- Different departments and network devices require appropriate traffic separation.
- The addition of four CCTV cameras requires the network to accommodate and segment CCTV traffic.
- Network redundancy is needed to support the assigned STP challenge and prevent switching loops.
- The network requires an organised IP addressing structure based on the assigned *10.12.0.0/16* address block.

Design Needs:

- Provide reliable connectivity between the required users, devices and network services.
- Implement VLAN segmentation for departments, records, CCTV and network management.
- Configure proper access controls for the confidential Records network.
- Configure and demonstrate **STP** for loop prevention and root bridge design.
- Provide connectivity for the four CCTV cameras introduced through **CR7** while keeping their traffic segmented.
- Provide the required routing and switching functionality in Cisco Packet Tracer.
- Design the network with sufficient capacity for future users, devices and departments.

- Ensure the completed network can be tested and show successful end-to-end connectivity.

1.3 User Needs and Departments

Based on the healthcare environment and the required network functionality, the following user groups and network segments are proposed.

User/Department	Network Needs	Proposed VLAN
Administration	Administrative applications, billing and Internet access	VLAN 10
Doctors	Access to required customer/medical records and Internet	VLAN 20
Reception	Patient check-in, scheduling and Internet access	VLAN 30
Records	Customer records and records-management systems	VLAN 40
CCTV	Connectivity for four CCTV cameras	VLAN 50
Network Administration	Management of network devices	VLAN 99

The departments are separated logically using VLANs. This reduces unnecessary broadcast traffic and provides better control over communication between different groups.

1.4 Applications and Services

The proposed network must support the following applications and services:

Application/Service	Users	Network Requirement
Customer/Medical Records	Doctors and Records	Controlled and reliable access
Appointment Scheduling	Reception	Reliable connectivity
Billing/Administrative Applications	Administration	Reliable connectivity
Office Applications	Staff	Network connectivity
CCTV Monitoring	Authorised users	Segmented CCTV traffic
Internet Access	Relevant users	External connectivity
Network Management	Network Administrators	Controlled management access

1.5 Security Requirements

The main security requirement specified by the client is that customer records are confidential and that access must be controlled.

To address this requirement, the Records department is placed on a separate VLAN. Access controls can then be applied to control which users are allowed to communicate with the Records network.

Security Requirement	Design Response
Confidential customer records	Dedicated Records VLAN
Controlled access	Access controls between relevant networks
Departmental separation	VLAN segmentation
CCTV traffic separation	Dedicated CCTV VLAN
Network device management	Dedicated Management VLAN
Unauthorised switch access	Appropriate switch security configuration

VLAN segmentation is used as part of the security design, but access controls are also needed because VLANs alone do not determine which users are allowed to access confidential resources.

1.6 Budget and Equipment Considerations

The project brief does not provide a specific monetary budget. Therefore, the equipment is selected based on the required network functionality while avoiding unnecessary network infrastructure.

Equipment	Quantity	Purpose
Cisco 2911 Router	1	Routing and external connectivity
Cisco 3560-24PS Switch	1	Core/distribution switching and STP root
Cisco 2960-24TT Switch	2	Access-layer connectivity
CCTV Cameras	4	CR7 requirement
End Devices	As required	Departmental connectivity
Network Management PC	1	Network administration

The selected devices provide the required routing, switching, VLAN and STP functionality needed for the proposed network.

1.7 Future Growth

The assigned 10.12.0.0/16 address block provides sufficient address space for future network expansion.

The proposed VLANs use /24 subnets, with each subnet providing 254 usable host addresses.

Future growth can be accommodated through:

- Adding more users to existing VLANs.
- Creating additional VLANs for new departments.
- Adding more CCTV cameras to the CCTV network.
- Adding more medical or network devices.
- Allocating additional /24 networks from the assigned /16 address block.

This allows the network to grow without requiring a complete redesign of the addressing scheme.

1.8 Requirements-to-Design Analysis

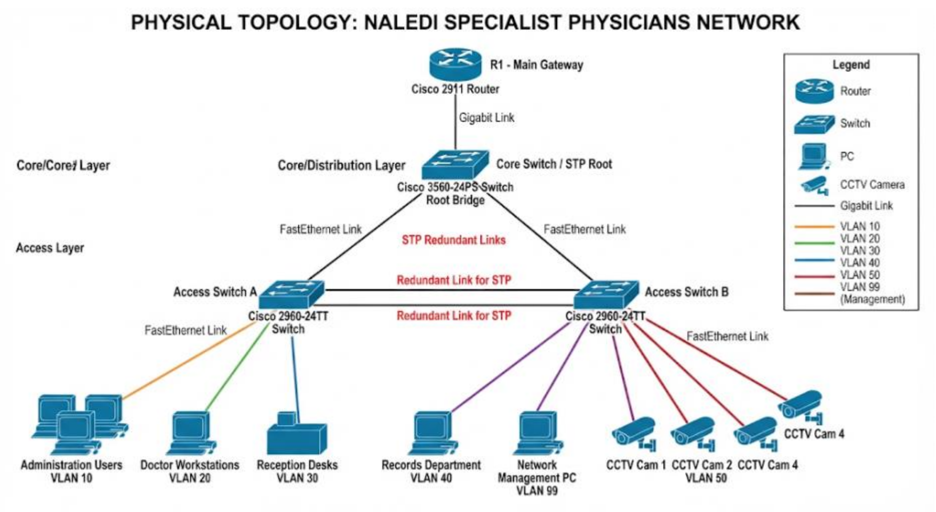
The client requirements were translated into the following network design decisions.

Client Requirement	Identified Need	Design Decision
Healthcare environment	Sensitive customer information	Network segmentation and controlled access
Confidential customer records	Records must be protected	Records VLAN and access controls
Appropriate connectivity	Required nodes must communicate	Router, switches and appropriate VLAN connectivity
STP challenge	Loop prevention and root design	Redundant Layer 2 connections and STP
CR7	Four CCTV cameras must be added	Dedicated CCTV VLAN
Addressing requirement	Network must use assigned address block	10.12.0.0/16 addressing plan
Future expansion	Network must accommodate growth	/24 subnets and additional VLAN capacity
Network management	Network infrastructure must be manageable	Dedicated Management VLAN

This analysis ensures that the physical and logical network designs are directly linked to the client's requirements.

2. Physical Topology

2.1 Physical Topology Diagram



The physical topology consists of:

1 × Cisco 2911 Router

1 × Cisco 3560-24PS Switch

2 × Cisco 2960-24TT Switches

Departmental end devices

4 × CCTV cameras

Network management device

Redundant switching connections for STP

2.2 Physical Design Overview

The physical topology uses a hierarchical arrangement consisting of a Cisco 2911 router, a Cisco 3560-24PS core/distribution switch and two Cisco 2960-24TT access switches. The router provides connectivity between the internal network and the external network, while the 3560 switch provides the central switching function.

The two 2960 switches provide connectivity for the different departments and network devices. Redundant Layer 2 connections are included to support the assigned STP challenge. STP will be configured to prevent switching loops and to establish a controlled root bridge while maintaining an alternative path.

The four CCTV cameras required by CR7 are connected to the access layer and will be assigned to a dedicated CCTV VLAN. This separates CCTV traffic from the departmental networks.

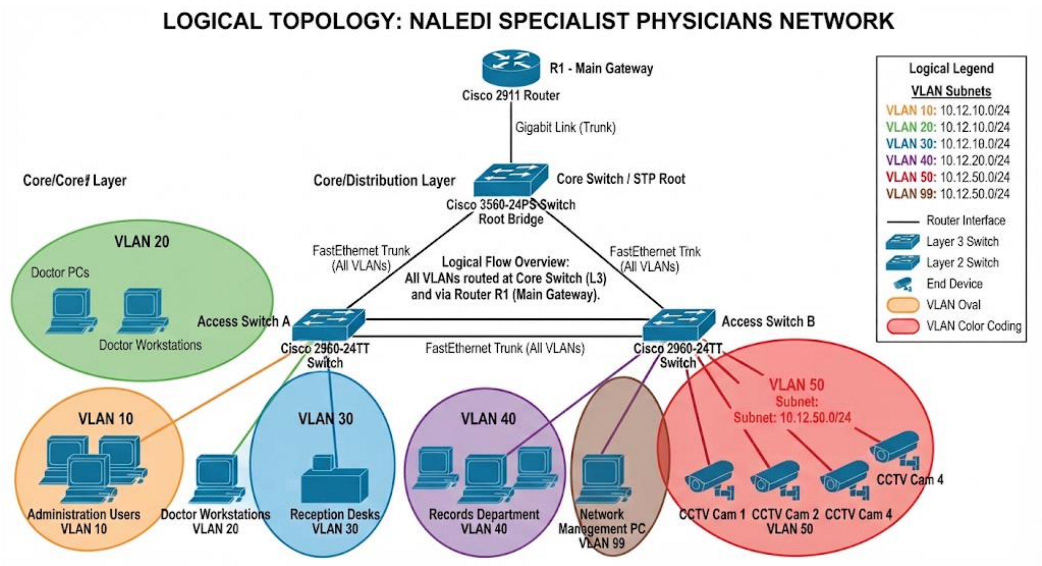
The physical design provides structured connectivity, redundancy and traffic segmentation while remaining appropriate for the requirements of Naledi Specialist Physicians.

2.3 Physical Design Decisions:

- The Cisco 3560-24PS is used as the central/core switch because it provides an appropriate location for the STP root design.
- Two Cisco 2960-24TT switches are used at the access layer to connect departmental devices and other network nodes.
- Redundant links are included between the switches because STP is the assigned networking challenge. These links create an alternative path while STP prevents a Layer 2 switching loop.
- The four CCTV cameras are connected to the access layer and will be placed in a separate CCTV VLAN to satisfy CR7.
- The router provides connectivity between the internal network and the external network.

3. Logical Topology

3.1 Logical Topology Diagram



The logical topology will show the VLAN structure and communication between the different network segments.

VLAN	**Name**	**Purpose**
10	Administration	Administration users
20	Doctors	Doctors and clinical users
30	Reception	Reception users
40	Records	Confidential customer records
50	CCTV	Four CCTV cameras
99	Management	Network management

VLAN segmentation separates users and services according to their network requirements. VLAN 40 provides logical separation for confidential records, while VLAN 50 ensures that CCTV traffic introduced through CR7 is separated from normal departmental traffic.

VLAN 99 is used for network management so that network infrastructure management traffic is separated from normal user traffic.

3.3 Logical Design Decisions

- VLANs are used to separate departments and different types of network traffic. This reduces the size of broadcast domains and provides a basis for controlling communication between different parts of the network.
- The Records VLAN is separated because customer records are confidential and access must be controlled.
- The CCTV VLAN is separated to satisfy CR7 and prevent CCTV traffic from being mixed with normal departmental traffic
- Trunk connections will be used where multiple VLANs need to pass between switches and the routing device.

- STP will be configured on the switching infrastructure to prevent Layer 2 loops and establish the appropriate root switch.

4. IP Addressing Plan

4.1 Addressing Scheme

- The assigned network block is 10.12.0.0/16
- The network is divided into /24 subnets to provide a simple and organised addressing structure.
- Each /24 subnet provides 256 total addresses and254 usable host addresses

4.2 VLAN and Subnet Allocation

VLAN	Department/Function	Network Address	Default Gateway	Subnet Mask
10	Administration	10.12.10.0/24	10.12.10.1	255.255.255.0
20	Doctors	10.12.20.0/24	10.12.20.1	255.255.255.0
30	Reception	10.12.30.0/24	10.12.30.1	255.255.255.0
40	Records	10.12.40.0/24	10.12.40.1	255.255.255.0
50	CCTV	10.12.50.0/24	10.12.50.1	255.255.255.0
99	Management	10.12.99.0/24	10.12.99.1	255.255.255.0

4.3 CCTV IP Address Allocation

Device	VLAN	IP Address
CCTV Camera 1	50	10.12.50.10
CCTV Camera 2	50	10.12.50.11
CCTV Camera 3	50	10.12.50.12
CCTV Camera 4	50	10.12.50.13

The four CCTV cameras are assigned addresses from the dedicated VLAN 50 subnet. This provides the required traffic segmentation for CR7.